Lei Zhang

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Research Interests

- *Multimodal Intelligence* — Advancing both the native capabilities and agentic wisdom of multimodal models to empower them with more sophisticated reasoning abilities in understanding and generation.
- *Scalable Training Methodology* — Developing data curation, evaluation, and training strategies to scale multimodal models toward stronger and more generalizable intelligence.
- *Agent Harness* - Exploring how agent harness can further extend the capabilities of frontier models through recursive self-improvement and customize environment of open-sourced model reinforcement learning.

Experiences

2026 – Now Research Intern, Nvidia Research
Host: Hanrong Ye, Pavlo Molchanov

2025 Research Scientist Intern, Meta GenAI
Host: Zecheng He, Junjiao Tian, Zhipeng Fan

Education

2024 – Now Ph.D., Computer Science, University of California San Diego
Advisor: Prof. Julian McAuley

2024 M.S., Computer Science, Zhejiang University, China

2021 B.S., Software Engineering, South China University of Technology, China

Awards & Honors

2025 Notable Reviewer Award, ICLR 2025

2023 Xiaomi Scholarship (10,000 RMB)

Publications

$*$ → equal contribution † → core contributor # → corrsponding author

Google Scholar Profile

Preprints

[1] Yuchen Lu, Julian McAuley, **Lei Zhang**[#].

Pack Light, CarryOn: Building Object-Oriented Context for Long-Horizon Agents **(2026)**. PDF

[2] Wei Chow, **Lei Zhang**, Xiongwei Zhu, XiuYu Zhang, Shuyue Dong, Tian Ye, Xian Sun, Yingshuo Wang, Linfeng Li, Lingdong Kong, Leigang Qu, Julian McAuley.
Generalizing On-Policy Self-Distillation for Diffusion Models **(2026)**.

[3] Wei Zhou, Xiongwei Zhu, Lingdong Kong, Bo Chen, **Lei Zhang**, Yongyuan Liang, Xiaoxia Hou, Ye Tian, Xian Sun, Yingshuo Wang, Linfeng Li, Shengqiong Wu, Leigang Qu, Feng Li, Wei Liu, Julian McAuley, Tat-Seng Chua.
On-Policy Self-Distillation in Diffusion Models **(2026)**. PDF
Featured by ByteDance Seed

[4] Fangxun Shu, **Lei Zhang***, Yongjie Ye, Yue Liao, Haiyang Guo, Gengyuan Liu, Liang Xiao, Shuicheng Yan, Chao Feng.
Variance-aware Sequence Policy Optimization **(2026)**.

[5] Yiwei Yang, Haoxiang Zhang, Bingbing Wen, Yao Lu, Yuchen Wu, **Lei Zhang**, Julian McAuley, Pan Lu, Bill Howe.
Spurious Tool Use: When RL Agents Learn the Wrong Reason to Act **(2026)**.PDF

Refereed Papers

[1] **Lei Zhang**, Junjiao Tian, Zhipeng Fan, Kunpeng Li, Jialiang Wang, Weifeng Chen, Markos Georgopoulos, Felix Juefei-Xu, Manling Li, Julian McAuley, Zecheng He.
Think in Strokes, Not Pixels: Process-driven Image Generation via Interleaved Reasoning
European Conference on Computer Vision (**ECCV 2026**). PDF
#1 Huggingface Daily Paper
Featured by DeepLearning.AI in *The Batch* and Meta AI

[2] Haoxiang Zhang†, Qixin Xu†, Zhuofeng Li†, **Lei Zhang**†, Pengcheng Jiang, Caiming Xiong, Yu Zhang, Julian McAuley.
Masking Stale Obeservations Helps Search Agents - Until It Doesn't: A Regime Map and Its Mechanism. 2026.
Empirical Methods in Natural Language Processing (**EMNLP 2026**). PDF

[3] Fangxun Shu†, Yue Liao†, **Lei Zhang**†, Le Zhuo, Chenning Xu, Guanghao Zhang, Haonan Shi, Weilong Dai, Tao Zhong, Zhelun Yu, Wanggui He, Siming Fu, Haoyuan Li, Hongsheng Li, Hao Jiang.
LLaVA-MOD: Making LLaVA Tiny Via MOE-Knowledge Distillation
International Conference on Learning Representations (**ICLR 2025**). PDF

[4] Sucheng Ren, Xianhang Li, Haoqin Tu, Feng Wang, Fangxun Shu, **Lei Zhang**, Jieru Mei, Linjie Yang, Peng Wang, Heng Wang, Alan Yuille, Cihang Xie.
Autoregressive pretraining with mamba in vision
International Conference on Learning Representations (**ICLR 2025**). PDF

[5] Wanggui He, Siming Fu, Mushui Liu, Xierui Wang, Wenyi Xiao, Fangxun Shu, Yi Wang, **Lei Zhang**, Zhelun Yu, Haoyuan Li, Ziwei Huang, Leilei Gan, Hao Jiang.
MARS: Mixture of Auto-Regressive Models for Fine-grained Text-to-image Synthesis
Association for the Advancement of Artificial Intelligence (**AAAI 2025**). PDF

[6] Jiaqi Chen, Xiaoye Zhu, Tianyang Liu, Ying Chen, Xinhui Chen, Yiwen Yuan, Chak

Tou Leong, Zuchao Li, Tang Long, **Lei Zhang**, Chenyu Yan, Guanghao Mei, Jie Zhang, Lefei Zhang.
Imitate Before Detect: Aligning Machine Stylistic Preference for Machine-Revised Text Detection
Association for the Advancement of Artificial Intelligence (**AAAI 2025**). PDF
Oral (4.6%)

[7] **Lei Zhang**, Jie Zhang, Bowen Lei, Subhabrata Mukherjee, Xiang Pan, Bo Zhao, Caiwen Ding, Yao Li, Dongkuan Xu.
Accelerating Dataset Distillation via Model Augmentation
IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2023**). PDF
Highlight (2.5%)

[8] **Lei Zhang**, Zhibo Wang, Xiaowei Dong, Yunhe Feng, Xiaoyi Pang, Zhifei Zhang, Kui Ren.
Towards Fairness-aware Adversarial Network Pruning
IEEE International Conference on Computer Vision (**ICCV 2023**). PDF
Featured by Adobe Research

[9] Jie Zhang*, **Lei Zhang***, Gang Li, Chao Wu.
Adversarial Examples for Good: Adversarial Examples Guided Imbalanced Learning
IEEE International Conference on Image Processing (**ICIP 2022**). PDF

[10] Jie Zhang, Bo Li, Jianghe Xu, Shuang Wu, Shouhong Ding, **Lei Zhang**, Chao Wu.
Towards Efficient Data Free Black-box Adversarial Attack
IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2022**). PDF

Technical Reports

[1] **Lei Zhang**, Fangxun Shu, Tianyang Liu, Sucheng Ren, Hao Jiang, Cihang Xie.
Filter & Align: Leveraging Human Knowledge to Curate Image-Text Data. 2024. PDF

[2] Fangxun Shu*, **Lei Zhang***, Hao Jiang, Cihang Xie.
Audio-visual LLM for Video Understanding. 2024. PDF

[3] Wenqiao Zhang, Tianwei Lin, Jiang Liu, Fangxun Shu, Haoyuan Li, **Lei Zhang**, Wanggui He, Hao Zhou, Zheqi Lv, Hao Jiang, Juncheng Li, Siliang Tang, Yueting Zhuang.
HyperLLaVA: Dynamic Visual and Language Expert Tuning for Multimodal Large Language Models. 2024. PDF

Professional Activities & Service

Reviewer, CVPR, NeurIPS, ICLR, ECCV, ACL

Notable reviewer, ICLR 2025

Teaching

Spring 2026 **Teaching Assistant**, Web Mining and Recommender Systems, UCSD

Fall 2025 **Teaching Assistant**, Maching Learning for Music, UCSD